

What’s the Catch?

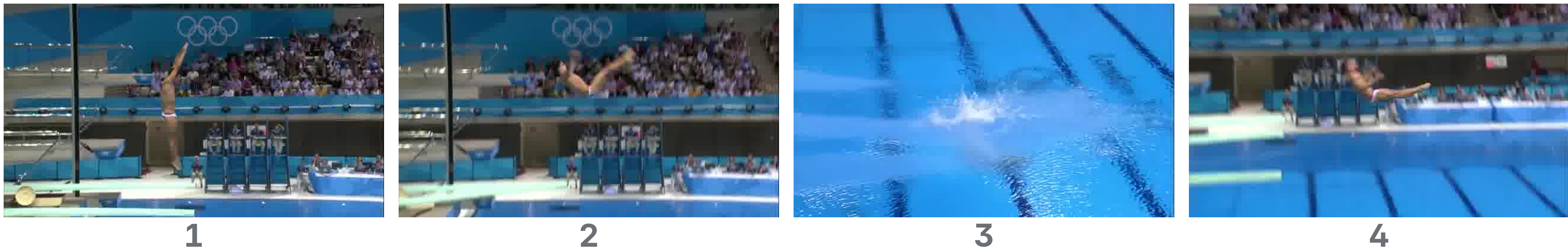
Evaluating Temporal Consistency in Vision-Language Models

Vision-language models can spot a corrupted frame — but not a reordered one.

Anonymous ACL submission

1 Try it — can you spot the swap?

Four frames from one Olympic dive. **Two adjacent frames have been swapped.** Which two?



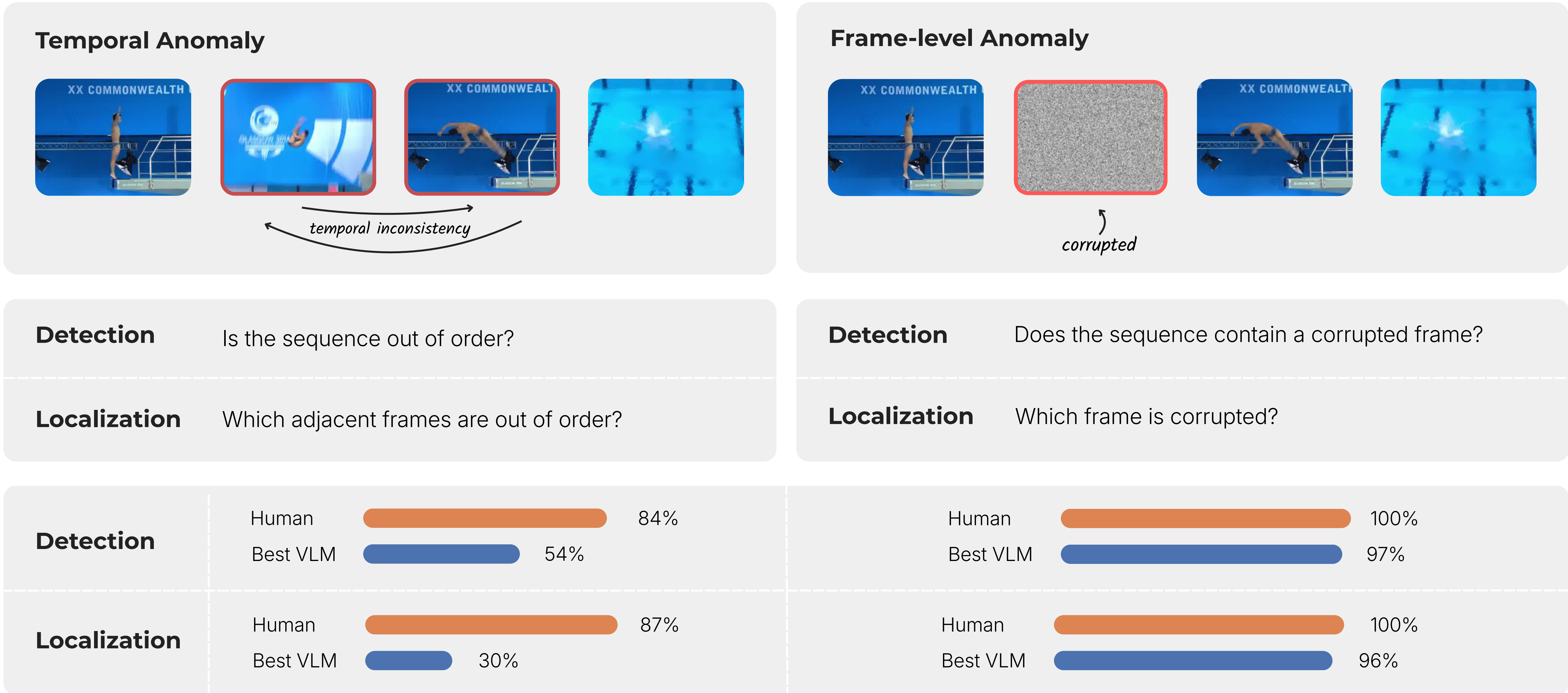
2 What we test, and why it is hard to measure

**The question.** Does a VLM connect information *across* frames, or does it only read each frame in isolation?

**The problem.** Many video benchmarks can be solved from a single frame or from language shortcuts — a high score does not prove temporal understanding.

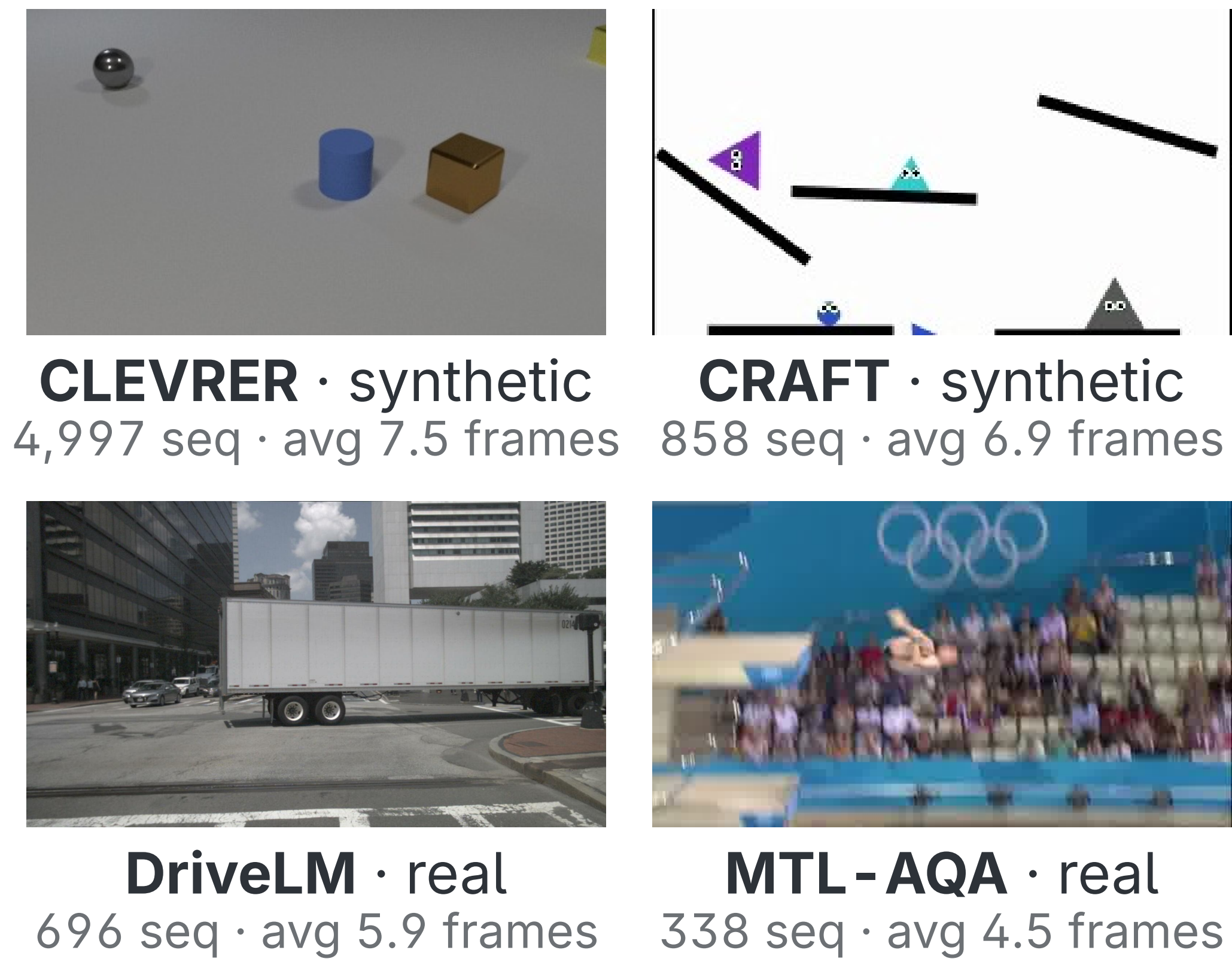
**The stakes.** Driving, robotics, and video understanding depend on recognizing how a scene *evolves*, not just what is in one frame.

3 The catch: models see broken frames, not broken time



Models detect and localize a corrupted frame reliably. They cannot detect or localize a reordered one. Humans do both near-perfectly.

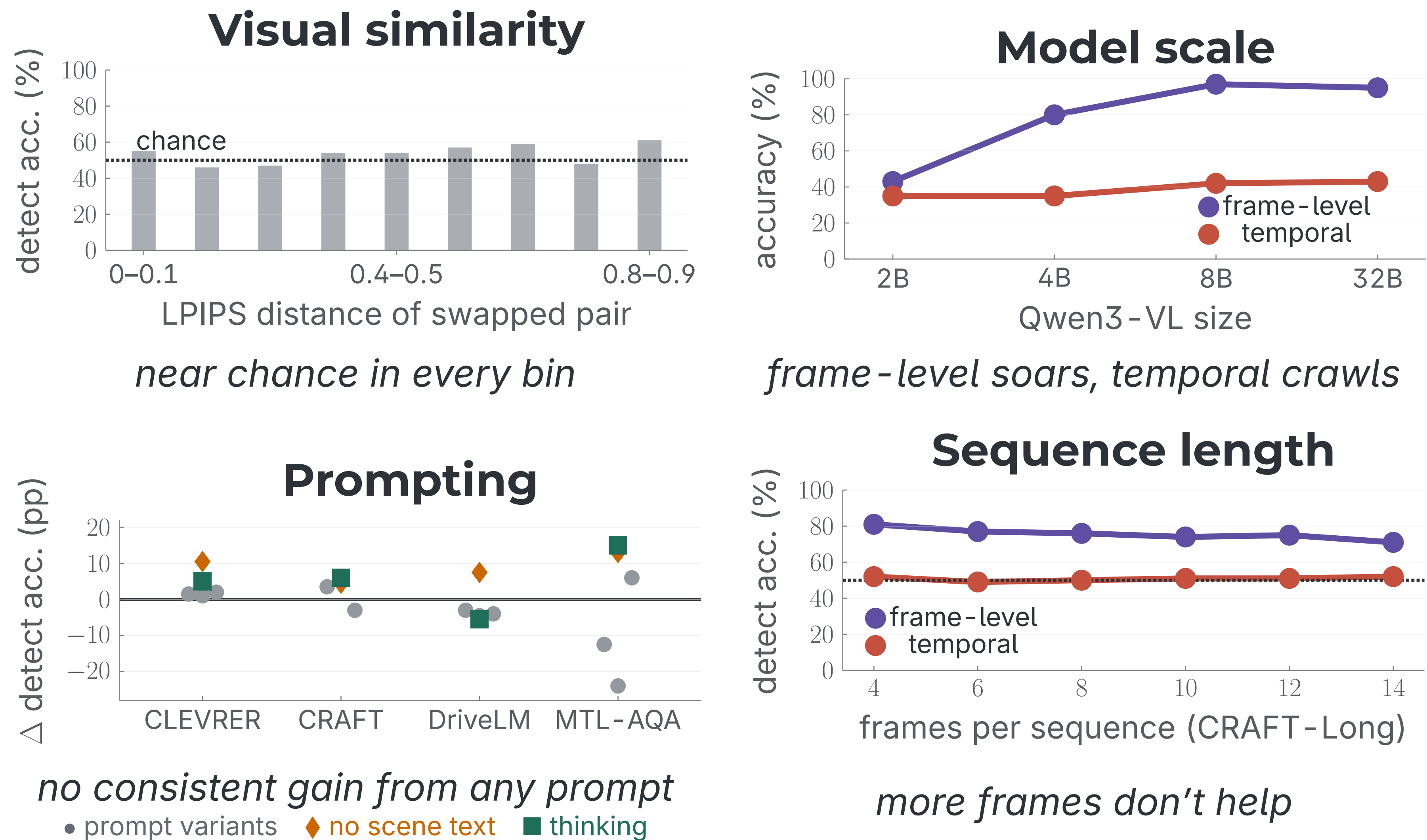
4 The benchmark



Near-identical consecutive frames are filtered out (LPIPS < 0.05), so every swap is **perceptually visible** — the anomaly is never hidden in lookalike frames.

5 What we ruled out

None of these explain the gap — the limitation looks **structural, not incidental.**



6 Takeaway

Current VLMs reason well about **individual frames** but not about how frames **relate over time**. TimeCatch isolates this by changing *only* frame order — everything else about the sequence stays the same. Scaling models up or changing prompts does not close the gap.